Qi Sun

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Research Interests: Teaching small language models to act as reliable search agents, through on-policy and offline distillation and model merging, and validating retrieval systems in clinical settings.

EDUCATION

New York University Shanghai Shanghai, China
B.S. in Computer Science Expected May 2027
Capstone (solo): *Right Finding, Wrong Organ: Auditing Anatomical Scope in Clinical Report Retrieval*

PUBLICATIONS

[1] **Qi Sun**[*], Siyue Zhang[*†], Yulin Chen, Yuxiang Xue, Ru Peng, Chen Zhao[†]. "From 'Weak' Signals to Strong Models: Preference Delta Aggregation with LoRA Merging." **NeurIPS 2026.** arXiv:2606.00357.

[2] Yizhou Liu[*], **Qi Sun**[*], Yulin Chen, Siyue Zhang, Chen Zhao. "Search, Do not Guess: Teaching Small Language Models to Be Effective Search Agents." *Committed to* **EACL 2027.** arXiv:2604.04651.

[3] **Qi Sun**[*], Siyue Zhang[*†], Tingyu Song, Yilun Zhao, Chen Zhao[†]. "Rethinking Offline On-Policy Distillation: From Scalar Advantages to Distributional Supervision." *Under review at ACL Rolling Review (October 2026).*

[*]Equal contribution. [†]Corresponding author.

RESEARCH EXPERIENCE

Research Assistant 2026.05 – Present
Advisor: Prof. Yiqiu (Artie) Shen NYU Langone Health, New York, NY

Feature-oriented retrieval over radiology reports for early pancreatic cancer (PDAC) detection

- Built a retrieval system over abdominal CT/MRI radiology reports and structured EHR that surfaces diagnostically similar cases for ambiguous pancreatic presentations.
- Designed a clinical codebook that encodes radiographic findings, quantitative thresholds, and negation as structured features for a hybrid BM25 and dense retrieval index, and generalized it into a disease-agnostic configuration reusable across conditions.
- Integrated retrieval with internal LLMs to estimate malignancy probability and generate evidence-grounded differential diagnoses. Manuscript in preparation for a medical informatics journal.

Undergraduate Researcher 2024.06 – Present
Advisor: Prof. Chen Zhao NYU Shanghai

Distributional on-policy distillation for agentic search (DistOPD) [3] 2026.06 – Present

- Showed that offline on-policy distillation (Lightning OPD), which caches teacher log-probabilities of recorded tokens on fixed reference rollouts, improves math reasoning but consistently underperforms its SFT starting point in multi-turn agentic search.
- Traced the failure to a larger mismatch between teacher and reference than in math and to distributional flattening after training, and showed theoretically that scalar-advantage supervision on fixed rollouts admits flattened solutions.
- Proposed DistOPD, which keeps candidate-token probabilities from both the reference and the teacher at selected positions and aggregates the rest into a residual token, so the teacher is queried once and never during training.
- DistOPD matches online OPD on agentic search with 10.6× fewer training GPU-hours, at a cost comparable to Lightning OPD. First author (equal contribution).

Preference Delta Aggregation for weak-to-strong training (PDA) [1] 2026.01 – 2026.05

- Proposed PDA, which turns each pair of a weak and a weaker model (e.g., Qwen3-4B over 1.7B) into a preference delta, trains it as a LoRA adapter, and aggregates many such deltas by LoRA merging.
- Introduced Geometric Alignment Merging (GAM), which aligns adapter subspaces to reduce directional interference between merged deltas.
- Improved a strong Qwen3-8B model by 6.8 / 7.3 points on knowledge reasoning and agentic search benchmarks, beating the best single-delta baseline by 2.1 / 4.3 points.
- First author (equal contribution) and presenting author. Accepted at NeurIPS 2026.

Small language models as effective search agents [2] 2025.06 – 2025.12

- Distilled agentic search behavior from Qwen3-32B into Qwen3-1.7B with SFT and on-policy distillation, against reproduced Search-o1 and Search-R1 baselines.
- Designed probing and rejection fine-tuning to teach adaptive when-to-search behavior, improving small-model search on multi-hop reasoning tasks.
- Co-first author. Committed to EACL 2027. Supported by the Dean's Undergraduate Summer Research Fund.

Earlier projects 2024.06 – 2025.05

- **Fact-checking with "Not Enough Information" labels.** Analyzed NEI label distribution and dataset shift across VitaminC and SciFact, with error analysis and fine-tuned baselines for ambiguous evidence.
- **Financial report summarization.** Summarized 10-K and 10-Q filings with BART, LLaMA, and GPT models, and traced summaries back to source text with MPNet-based retrieval.

HONORS & AWARDS

- Dean's Undergraduate Summer Research Fund, NYU Shanghai 2025
- ICPC Regional Contest (Ningxia), Bronze Prize 2025
- CSAW LLM CTF 2024 (MENA), Second Place 2024

LEADERSHIP

- **Chair,** NYU Shanghai ACM Student Chapter 2024.06 – 2025.06
- **Director of Competition,** NYU Shanghai ICPC Training Community 2024.06 – 2025.06
  With Prof. Xianbin Gu, built the team's training platform (Vue3, Django) with an online judge and LLM-based personalized problem recommendation.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, SQL, JavaScript
- **ML Stack:** PyTorch, Hugging Face Transformers, PEFT/LoRA, vLLM, FAISS, Slurm multi-GPU clusters